STANDARD OPERATING PROCEDURE (SOP)

Adverse Pregnancy Outcome Clinical Audit & BMO Feedback Dashboard

Operational Protocol for District Hospital Khargone & Peripheral Health Blocks (Madhya Pradesh)

Document Version: 1.0 (Final)	**Target District:** West Nimar (Khargone), M.P.
Author / Unit: District QA & Sankalp Project Team	**Nodal Facility:** District Hospital (DH) Khargone
Target Audience: BMOs, MO I/Cs, CHOs, ANMs, BPMs, M&E;	**Effective Date:** September 2026 (Monthly Cycle)
Outputs: Interactive HTML & 11-Tab Styled Excel Linelist	**Mandate:** Zero Missed High-Risk ANC in Peripheral Care

EXECUTIVE MANDATE & OPERATIONAL CONTEXT:
Every month, pregnant women present to District Hospital Khargone with severe adverse pregnancy complications (Severe Preeclampsia, Eclampsia, Hb < 7.0 g/dL, Sickle Cell Crisis, and Perinatal Death). An audit of clinical records reveals that over 70% of these adverse outcomes stem from missed screening, delayed risk identification, or absent pre-referral stabilization during peripheral Antenatal Care (ANC) at Sub-Health Centres (SHC), Health & Wellness Centres (HWC), and Primary Health Centres (PHC).

This SOP establishes the complete standard operating procedure for generating the **BMO Adverse Outcome Feedback Dashboard** and **11-Tab Styled Excel Linelist**. It enables the Chief Medical & Health Officer (CMHO) and Block Medical Officers (BMOs) to trace each tertiary complication back to its assigned Subcentre and frontline workers (ANM, CHO, ASHA) and enforce corrective actions.

1. Clinical Definitions & Analytical Framework

This SOP establishes precise clinical definitions, diagnostic thresholds, algorithmic detection rules, and peripheral care preventative imperatives used in the audit system:

Clinical Condition & Threshold	Detection Rule (Regex)	Diagnostic Criteria & ANC Quality Gap Rationale
Severe Preeclampsia / PIH BP ≥ 160/110 mmHg on 2 occasions with proteinuria (≥ 2+) after 20 weeks, or gestational hypertension with end-organ dysfunction (headache, vision blur, epigastric pain).	`preeclampsia|pre eclampsia|pih`	Reflects missed weekly BP screening at HWC/PHC and absence of oral anti-hypertensives (Labetalol / Methyldopa) in peripheral care.
Hypertensive Disorders (Eclampsia) Generalized tonic-clonic convulsions or coma in a preeclamptic woman without other neurological causes.	`convulsion|(?<!pre\s)(?<!pre)eclampsia`	Demonstrates absence of pre-referral stabilization: **failure to administer loading dose of Magnesium Sulfate (Inj. MgSO4 4g IV + 10g IM)** before ambulance transfer.
Severe Anemia Hemoglobin level < 7.0 g/dL during pregnancy (critical values < 4.0 g/dL).	`anemia|anaemia`	Reflects missed early registration, unverified IFA consumption, or failure to administer **IV Iron Sucrose or Inj. FCM** in 2nd/3rd trimester at CHC/PHC.
Sickle Cell Disease Crisis Vaso-occlusive pain crisis, acute chest syndrome, or severe hemolytic crisis in HbSS or HbS-β thalassemia patients.	`scd|sickle`	Prevalent tribal complication. Indicates missed universal screening under MP Sickle Cell Mission, lack of 5mg daily Folic Acid, and unmonitored dehydration.
Perinatal Loss (Stillbirth / IUFD) Fetal demise at ≥ 28 weeks of gestation (or birth weight ≥ 1000g).	`iud|iufd|stillbirth|msb`	Direct indicator of missed fetal heart sound monitoring, unaddressed IUGR, severe oligohydramnios, or undetected fetal distress.
Early Neonatal Death Death of a live-born infant within 7 completed days of life (0–168 hours).	`died|death in baby status`	Attributable to severe intrapartum birth asphyxia, meconium aspiration, hypothermia during transport, or extreme prematurity.
SNCU Transfer / Admission Transfer of neonate to Special Newborn Care Unit within hours of delivery.	`yes|shift|lbw|gasping|msl`	Primary reasons: Meconium Stained Liquor (MSL), birth weight < 1800g, respiratory distress, prematurity, or sepsis.
Maternal Death & Near-Miss Maternal death during pregnancy/puerperium; or near-fatal collapse requiring tertiary transfer (MY Hospital Indore).	`died|death|refer|indore`	Terminal indicators of delayed danger sign recognition (Delay 1), transport delay (Delay 2), or delayed resuscitation (Delay 3).

HRP Flagged in ANMOL Whether patient was formally registered and flagged as High Risk on portal.	`Cross-match with HRP report`	Audits frontline tracking compliance. If a mother with severe complications was unflagged, it represents an administrative tracking failure.

THE THREE DELAYS AUDIT FRAMEWORK:
All adverse cases must be audited through the Three Delays model during monthly BMO reviews: • **Delay 1 (Care Seeking):** Family/mother delayed in recognizing danger signs (e.g. severe headache, blurred vision, absent fetal movement).
• **Delay 2 (Transport / Referral):** Delay in reaching the referral hospital due to ambulance availability, bad terrain, or multiple intermediate stops.
• **Delay 3 (Facility Protocol):** Delay in receiving timely, standardized clinical care at the facility (lack of MgSO4, IV iron, or blood products).

2. Data Sources & Complete Variable Dictionary

The audit pipeline ingests four foundational datasets. Below is the complete variable dictionary detailing every file, sheet name, column letter, column index, header name, data type, and its operational role:

2.1 Primary Adverse Outcome Register (DH Delivery Register)

Source File: Data/Adverse outcome linelist.xlsx | **Sheet:** Sheet1 | **Header Row:** 1 (Index 0)

Col	Index	Source Header Name	Data Type	Operational Role & Standardization Rule
A	0	**Sr No**	Integer	Case sequential identifier (1 to N). Primary audit tracking ID.
B	1	**MPID**	String/Float	RCH portal Maternal ID. Cleaned by stripping '.0', 'nan', 'direct entry'. Used as primary key for registry cross-matching.
C	2	**Admission date**	Datetime	Date of DH admission. Parsed into 'YYYY-MM-DD' and 'Month YYYY' for dynamic filtering.
D	3	**Patient Name**	String	Name of pregnant woman and husband (e.g. 'Chena bai/Indarsingh'). Trailing spaces trimmed.
E	4	**Address**	String	Stated village / locality address. Used in secondary fuzzy location triangulation.
F	5	**village**	String	Stated revenue village name.
G	6	**Sub centre**	String	Stated Subcentre / referring peripheral facility (e.g. 'phc oon', 'chc Bhagwanpura').
H	7	**Block**	String	Stated health block name. Used for initial block classification before hierarchy validation.
I	8	**Diagnosis**	String	Clinical diagnosis text recorded by DH obstetric team. Parsed via regex into 4 core categories.
J	9	**Mother status**	String	Maternal status at discharge ('live', 'died', 'refer indore'). Classified into Live / Maternal Death / Referral.
K	10	**Baby status**	String	Perinatal delivery outcome ('live', 'iud', 'died'). Classified into Live Birth / Stillbirth (IUFD) / Neonatal Death.
L	11	**Transfer of baby to sncu reason**	String	Indicates whether baby was transferred to SNCU and clinical justification (MSL, gasping, LBW, prematurity).
M	12	**mob num**	String/Float	Patient/family mobile number. Cleaned to 10 digits. Used as secondary cross-match key when MPID is missing.

2.2 ANC Line List Registry Report

Source File: Data/ANC_Line_list_Report_2026-09-17_09_45_52.xlsx | **Header Row:** 9 (skiprows=8) | **Total Columns:** 180

Col Header Name	Col Index	Data Type	Analytical Purpose & Matching Role in Dashboard
Health Block Name	3	String	Authoritative health block where patient is enrolled. Overrides inaccurate stated block.
Health Sub Centre Name	5	String	Authoritative Sub-Health Centre (SHC). Used to join with staff contact numbers.
Village Name	6	String	Enrolled revenue village name. Used for community audit tracing.
Name of ANM	7	String	Assigned Auxiliary Nurse Midwife responsible for antenatal tracking.
Name of ASHA	8	String	Assigned Accredited Social Health Activist for village home visits.
CHO Name	9	String	Assigned Community Health Officer at Health & Wellness Centre.
Name of Pregnant Woman	10	String	Enrolled pregnant woman name. Cross-verified with DH admission name.
MP ID	11	String	Primary matching key. Exact 11-digit maternal registration ID.
Mobile Number	17	String	Cleaned 10-digit mobile number. Secondary fallback matching key.
HRP Identified(Yes/No)	22	String	Whether patient was identified as High Risk Pregnancy during routine ANC.
Hb.1 - Hb.11 / BP.1 - .11	Various	Floats/Strings	Serial hemoglobin measurements and blood pressure checks to track clinical trajectory.

2.3 High Risk Pregnancy (HRP/UHRP) Line List Report

Source File: Data/HRP_Line_List_Report_2026-09-17_10_22_24.xlsx | **Header Row:** 8 (skiprows=7) | **Total Columns:** 76

Col Header Name	Col Index	Data Type	Analytical Role & Clinical Extraction Rule
Health Block Name	3	String	Authoritative health block name for confirmed high-risk cases.
Health Sub Centre Name	5	String	Authoritative subcentre for frontline staff linkage.
MP ID	11	String	Primary matching key to link adverse outcomes with confirmed high-risk registry.
Mobile Number	17	String	10-digit mobile fallback key for patient reconciliation.
Identified with PIH (Yes/No)	42	String	Identifies whether pregnancy-induced hypertension was flagged on the portal.
PIH Management Done	43	String	Records anti-hypertensive regimen administered (Methyldopa, Labetalol, Inj. MgSO4).
If severe anemia (Yes/No)	33	String	Flags clinical severe anemia enrollment on portal.
Inj FCM given (Yes/No)	40	String	Verifies whether IV Ferric Carboxymaltose was administered at facility.
Number of Blood transfusion units	39	String/Float	Quantity of packed red cells/whole blood transfused before delivery.
Result of sickle cell test	22	String	Result of solubility/electrophoresis test (Trait, Disease, Thalassemia, Negative).
GDM Positive / Management	44, 45	String	Gestational diabetes screening status and medical nutrition/insulin therapy.
Previous LSCS	29	String	Prior lower segment cesarean delivery. High risk indicator.

2.4 Master Location & Human Resources Hierarchy

Source File: Resources/Location Hirarchy.xlsx | Contains 4 official directories used for facility and personnel linkage:

Sheet Name	Header Row	Key Columns Used	Functional Integration in Pipeline
Village list	Row 2 (skip=1)	Col B: District Name Col D: Block Name Col G: Facility Name Col K: Subcentre Name Col O: Village Name	Authoritative location tree across all 9 blocks of Khargone. Used to resolve unbooked/unregistered cases by matching stated village names to official Subcentres and Blocks.
ANM	Row 3 (skip=2)	Col C: Block Name Col E: Sub-Health Facility Col F: ANM Name Col H: Contact No	Master roster of 339 Auxiliary Nurse Midwives. Joined on normalized Subcentre name (or ANM Name) to retrieve verified 10-digit mobile contact numbers.
CHO	Row 3 (skip=2)	Col C: Block Name Col E: Sub Health Facility Col F: Name of CHO Col H: Contact No	Master roster of 284 Community Health Officers at HWCs. Joined on normalized Subcentre name (or CHO Name) to retrieve verified mobile contact numbers.
HR Directory	Row 1 (skip=0)	Col B: Facility Name Col E: HR Name Col F: Designation Col I: Mobile Number	Secondary human resources verification file used when ANM or CHO contact numbers are missing or updated in field postings.

3. Multi-Tiered Data Triangulation & Matching Engine

To ensure zero patient leakage and complete spatial reconciliation, process_data.py implements a robust 6-tier hierarchical matching algorithm:

Matching Tier	Match Logic & Input Fields	Reconciliation Output & Impact
Tier 1: Universal MP ID Matching	Exact join on cleaned 11-digit MPID against ANC & HRP registries.	Highest confidence match. Instantly retrieves verified Health Block, Subcentre, Village, registered ANM/CHO/ASHA, and portal clinical tracking history.
Tier 2: Clean Mobile Phone Fallback	Join on stripped 10-digit mob num against ANC & HRP registries when MPID is missing or recorded as 'direct entry'.	Resolves direct hospital walk-ins who possessed portal registrations under their husband or family contact number.
Tier 3: Location Hierarchy Subcentre Join	Normalize verified Subcentre string (stripping 'SHC', 'PHC', 'New', punctuation, and spaces) and join against ANM & CHO sheets.	Pulls official ANM Name, ANM Contact Number, CHO Name, and CHO Contact Number directly into the patient linelist.
Tier 4: Personnel Name Fallback	Normalize ANM Name / CHO Name from registry and search anm_by_name / cho_by_name hash maps.	Retrieves active phone numbers even if subcentre naming conventions differ between portal dumps and HR directories.
Tier 5: Village Text Extraction (Unbooked Cases)	For unbooked walk-ins, extract geographic words from Address and village, join against Village list master.	Locates the exact administrative Subcentre and Block for mothers who bypassed peripheral care (e.g. Tharadpura → SHC Ambakheda in Bhagwanpura).
Tier 6: Out-of-District Classification	Identify geographic tokens outside Khargone: Sendhwa, Thikri, Rajpur, Nepanagar, Dahinala, Dharampuri, Punasa, Jalgaon.	Segregates external inflow cases into 'Out of District' block tagged by origin (Barwani, Burhanpur, Dhar, Khandwa, Maharashtra) to preserve Khargone block KPI integrity.

4. Technical Specifications of Deliverables

4.1 Interactive Web Dashboard (BMO_Adverse_Outcome_Dashboard.html)

- **Single Action Header:** Contains print trigger and strictly **1 global download button** (Download Excel Report) linking directly to the styled multi-tab workbook.
- **Prominent Filter Strip:** Instant client-side filtering by Block (All, Khargone Only, 9 individual blocks, Out of District), Admission Month, Clinical Complication, and Multi-field search.
- **Dynamic 6 KPI Cards:** Auto-recalculating metrics (Total Cases, Khargone Cases, Out of District, HDP/Eclampsia, Severe Anemia, Perinatal Loss). HRP enrolled card omitted as directed.
- **Exact Tab Order:** 1. Block Scorecard & Hierarchy → 2. Patient Linelist → 3. Out of District Inflow → 4. Overview & Gaps (Three Delays chart omitted) → 5. BMO Action Guidelines.
- **Linelist Table Formatting:** Columns arranged as: *Assigned Block & Subcentre → Patient & ID → Date of Admission → DH Diagnosis → HRP flagged in ANMOL → Perinatal/Maternal Outcome → Frontline Workers (with click-to-call phone badges) → Action Review*. Click-to-sort enabled across all columns.
- **Case Modal Dossier:** Detailed patient clinical profile, high-risk flags, 3-column staff card with direct tel: dial links, and the 3 concise BMO Action Directives.

4.2 Formatted Excel Linelist (Adverse_Outcome_Patient_Linelist_Styled.xlsx)

Generated via openpyxl with strict layout and color styling identical to the web dashboard:

Component / Tab	Exact Tab Order & Column Structure	Color Profile & Hex Styling
Tab 1: Block Scorecard (1st Place)	Scorecard of all 9 Khargone blocks with columns: Block Name, Total Adverse, Severe Preeclamp, Eclampsia, Severe Anemia, Sickle cell crisis, Stillbirths, SNCU transfer, Neo Deaths, plus Total Khargone District row.	Header fill: Deep Indigo #1E1B4B, White bold text. Total row fill: Slate #E2E8F0, text #0F172A.

Tabs 2 to 10: Block Linelists (Separate Tab per Block)	Separate dedicated tab for each Khargone block: **Barwaha** (8), **Bhagwanpura** (10), **Bhikangaon** (4), **Gogaon** (4), **Jhiranya** (7), **Kasrawad** (1), **Khargone** (5), **Maheshwar** (2), **Segaon** (5). **Strictly 16 Columns (Col A to P):** Sr No, Assigned Block, Assigned Subcentre, Village/Area, Patient Name, Mobile Number, MP ID, Date of Admission, Admission Month, DH Diagnosis, Clinical Category, HRP flagged in ANMOL, HRP High Risk Category, Perinatal Outcome, Mother Status, SNCU Transfer. **All columns after SNCU Transfer (Q to W) removed.**	**Clinical Diagnosis Fills:** • Severe Preeclampsia: #FEF3C7 (Text #92400E) • Eclampsia: #FEE2E2 (Text #991B1B) • Severe Anemia: #EDE9FE (Text #5B21B6) • Sickle Cell Crisis: #FFEDD5 (Text #9A3412) **Outcomes & HRP Fills:** • Live Birth: #DCFCE7 (Text #166534) • Stillbirth: #F1F5F9 (Text #475569) • Neonatal Death: #FEE2E2 (Text #991B1B) • Maternal Death: #991B1B (Text #FFFFFF) • HRP Flagged: #ECFDF5 (Text #047857) • Not in HRP: #FFF1F2 (Text #BE123C) • SNCU Yes: #E0E7FF (Text #3730A3)
Tab 11: Out of District Inflow (Last Tab)	External referral admissions from Barwani, Dhar, Burhanpur, Khandwa, and Maharashtra (10 cases). Same standardized 16 columns (Col A to P).	Assigned block cell styled with Light Amber fill #FFFBEB and text #B45309.

5. Step-by-Step Monthly Operational SOP Workflow

District Health Teams, Hospital Data Officers, and BMOs must execute the following protocol between the 1st and 15th of every month:

Timeline & Phase	Responsible Officer	Standard Operating Procedure Instructions
Phase 1: 1st–3rd of Month Data Ingestion	DH Delivery Room In-charge & SNCU Data Entry Operator	1. Export monthly delivery and maternal admission registers from District Hospital Khargone. 2. Populate Data/Adverse outcome linelist.xlsx with mandatory fields: Sr No, MPID, Admission date, Patient Name, Address, village, Sub centre, Block, Diagnosis, Mother status, Baby status, Transfer of baby to sncu reason, and mob num. 3. Ensure clean 10-digit mobile numbers are recorded for direct walk-ins lacking MPID.
Phase 2: 4th–5th of Month Portal Dump Extraction	District Data Officer / NHM M&E Consultant	1. Log into ANMOL/RCH portal and download cumulative ANC Line list Report for the previous month (→ Data/ANC_Line_list_Report_YYYY-MM-DD.xlsx). 2. Download the High Risk Pregnancy Line list Report (→ Data/HRP_Line_List_Report_YYYY-MM-DD.xlsx). 3. Verify that Resources/Location Hirarchy.xlsx reflects the latest staff postings.
Phase 3: 6th of Month Automated Processing	District MIS Unit / Systems Analyst	Execute terminal processing commands in the project directory: python process_data.py → Reconciles MPIDs, cross-matches registries, matches Location Hierarchy staff, outputs bmo_dashboard_data.json. python generate_styled_excel.py → Produces 11-tab styled Excel workbook Adverse_Outcome_Patient_Linelist_Styled.xlsx. python scratch/update_html.py → Updates web dashboard HTML payload.
Phase 4: 7th–8th of Month Data Verification & QC	Civil Surgeon & District Quality Assurance Officer	1. Open BMO_Adverse_Outcome_Dashboard.html in any modern browser. 2. Validate that total admissions equal DH registry count (56 cases). 3. Check block distributions in Scorecard tab. 4. Click [Download Excel Report] to ensure workbook contains 11 tabs, Block Scorecard is in 1st place, and all columns after SNCU transfer (Q to W) are absent.
Phase 5: 9th–10th of Month BMO Packet Dispatch	Chief Medical & Health Officer (CMHO)	1. Email the interactive HTML dashboard to all 9 Block Medical Officers. 2. Attach the respective block's styled Excel sheet to each BMO. 3. Issue formal audit meeting notice for the mid-month sector review.
Phase 6: 15th of Month Monthly Block Audit Meeting	BMO, Sector Medical Officers & CHOs	Enforce the **3 Mandatory BMO Action Directives:** **1. Do community audit why condition was not resolved?** Conduct field visits with ASHA/ANM to identify care-seeking delays (Delay 1). **2. Ask in review meeting to MO and CHO why women presented with adverse outcome?** Table specific patient names from linelist. Query CHO/ANM on missed BP/Hb monitoring and lack of pre-referral stabilization (Delay 3). **3. If found any system gaps try to resolve.** Restock IV Iron Sucrose/FCM, MgSO4, functional BP instruments, and enforce universal Sickle Cell screening.

6. Document Approval & Governance Sign-off

Role / Governance Designation	Office / Institution	Signature & Official Seal
Civil Surgeon / Hospital Superintendent	District Hospital Khargone (M.P.)	_____
Chief Medical & Health Officer (CMHO)	Department of Health & Family Welfare, Khargone	_____
District Nodal Officer (Maternal Health / NHM)	National Health Mission, Khargone	_____